Key Probability Inequalities - Math 742

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Markov's Inequality

For a **non-negative** random variable $X \geq 0$ and $t > 0$,

$$\boxed{\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}}$$

Chebyshev's Inequality

For any random variable X with finite mean μ and variance σ^2,

$$\boxed{\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}} \quad \text{or} \quad \boxed{\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}}$$

Cauchy–Schwarz (for expectations)

For random variables X, Y with finite second moments,

$$\boxed{|\mathbb{E}[XY]|^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]} \quad \rightsquigarrow \quad |\mathrm{Cov}(X,Y)|^2 \leq \mathrm{Var}(X)\mathrm{Var}(Y)$$

Jensen's Inequality

If f is **convex**, then

$$\boxed{f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]}$$

If f is **strictly convex**, equality iff X is degenerate.

Crucial example: $f(t) = -\log t$ is strictly convex on $(0, \infty) \Rightarrow$

$$\boxed{p \log p + (1-p)\log(1-p) \ \leq \ \log 1 = 0 \qquad (\text{entropy } \ \leq 0)}$$

and more generally for the binomial log-likelihood:

$$\boxed{S \log p + (n-S)\log(1-p) \ \leq \ n \log(S/n) < 0 \quad (0 < S < n)}$$

$$S = \sum_{i=1}^{n} x_i$$

Hoeffding's Inequality (bounded r.v.s)

Let $X_1, \ldots, X_n$ be independent, $a_i \le X_i \le b_i$ a.s. Then for $\varepsilon > 0$,

$$\boxed{\mathbb{P}(|\bar{X}_n - \mathbb{E}[\bar{X}_n]| \ge \varepsilon) \le 2\exp\Big(-2n^2\varepsilon^2 \Big/ \sum (b_i - a_i)^2\Big)}$$

Special case – Bernoulli(p) ($X_i \in \{0, 1\}$):

$$\boxed{\mathbb{P}(|\bar{X}_n - p| \ge \varepsilon) \le 2\exp(-2n\varepsilon^2)}$$

Mills' Inequality (Gaussian tail)

For $Z \sim N(0,1)$ and $t > 0$,

$$\boxed{\frac{1}{\sqrt{2\pi}} \frac{e^{-t^2/2}}{t}\left(1 - \frac{1}{t^2}\right) < \mathbb{P}(Z > t) < \frac{1}{\sqrt{2\pi}} \frac{e^{-t^2/2}}{t}}$$

Math 742 – Spring 2026. These are the **only** inequalities you will ever need for proofs in this course.